

Chapter 12: Simple Machines

Textbook Exercise Solutions & Answers

1. Classify the following as a lever, a pulley and an inclined plane:

(A wedge, a needle, a staircase, a slide, the wheel of a flagpole, nutcrackers, scissors, an opener, an axe, a crane, a knife)

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Inclined plane	Pulley	Lever
Wedge	Wheel of flagpole	Nutcracker
Needle	Crane	Scissors
Staircase		Opener
Slide		
Axe		
Knife.		

2. Fill in the blanks using the proper word and complete the statements.

- (a) The in the centre, the on one side and the on the other side make a lever of the first order.

Fulcrum, load and effort

- (b) The in the centre, the on one side and the on the other side make a lever of the second order.

Load, fulcrum, effort

- (c) The in the centre, the on one side and the on the other side make a lever of the third order.

Effort, fulcrum, load

3. Which machines will you use to do the following work? Write their types.

(a) To remove the lid of a tin.

- Work: To remove the lid of a tin.
- Machine: Opener
- Machine: Lever of the second order

(b) To lift bricks to the top of a tall building

- Work: To lift bricks to the top of a tall building
- Machine: Crane
- Machine: Complex machine

(c) To cut vegetable

- Work: To cut vegetable
- Machine: Knife
- Machine: Wedge – Simple machine

(d) To draw water from a well

- Work: To draw water from a well
- Machine: Pulley
- Machine: Simple machine

(e) To hold a papad for roasting it.

- Work: To hold a papad for roasting it.
- Machine: A pair of tongs
- Machine: Lever of the third order

4. Write the answers to the following questions in your own words.

(a) What is meant by a simple machine?

1. Machines which have only one or two parts and a simple and easy structure are simple machines.
2. Example: the bottle opener, nail cutter.
3. Simple machines can be handled easily.
4. We use many simple machines in our day to day life.

(b) Mention the advantages of using a machine.

1. Machines, simple or complex, are used to get more work done in less time and with less effort.
2. Machines are used to accomplish certain tasks more easily.
3. Use of simple or complex machines depends upon the task to be carried out and the time and the efforts required to do it.

(c) What is meant by a complex machine?

1. Complex machines have many parts which carry out many processes for completing a task.
2. For this purpose the parts are joined to one another.
3. Therefore, these machines are called complex machines.
4. Some of the parts of such complex machines are actually simple machines.
5. The structure of complex machine is complicated.

(d) What is a lever? How is the order of the lever determined?

1. Lever is a simple machine made up of 3 parts: Load, effort and fulcrum.
2. Order of the lever are determined on the position of the effort, the fulcrum and the load.

5. Why is this so?

(a) Traveller's bags have wheels.

1. Combination of wheel and axle is a simple machine.
2. When the axle starts rotating due to force applied, the wheel fitted to it also starts rotating.
3. This makes easier for travellers to carry heavy load.
4. Instead of lifting heavy bag, to drawing it on wheels require less energy. Therefore, traveller's bag have wheels.

(b) Machines have to be maintained.

1. Parts of machines rub against one another when they are used.
2. Soiled, dusty parts create more friction. Some parts are affected by the weather, rust and corrosion.
3. Machines get damaged or become useless due to such wear and tear. Hence to avoid this, machines should be maintained properly to keep them in good working condition.

(c) A bicycle is said to be a complex machine.

1. Bicycle has a complicated structure.
2. Bicycle has many parts which carry out many processes for completing a task.
3. Many parts of bicycle such as pedals, wheels, axle, handle are actually simple machines.
4. The parts of a bicycle are joined to one another – making it a complex machine.

6. Name the levers mentioned in the following passage. Identify the fulcrum, load and effort of each and say which type of lever it is.

Ravi and Savita sit on a sea-saw in a garden. In the mean time, a gardener is trimming trees in the garden. He puts the leaves and other garbage in the wheelbarrow. Later, Ravi gets thirsty and he buys lemon sherbet. The sherbet seller cuts the lemon and squeezes it using a lemon squeezer. He puts small pieces of ice in the glass with the help of the tongs.

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Lever:

1. See – saw – lever of the first order.
Load – fulcrum – effort.
2. Wheelbarrow – lever of the second order.
Fulcrum – load – effort.
3. Lemon squeezers – lever of the second order.
Fulcrum – load – effort.
4. Tongs – lever of the third order.
Fulcrum – effort – load.